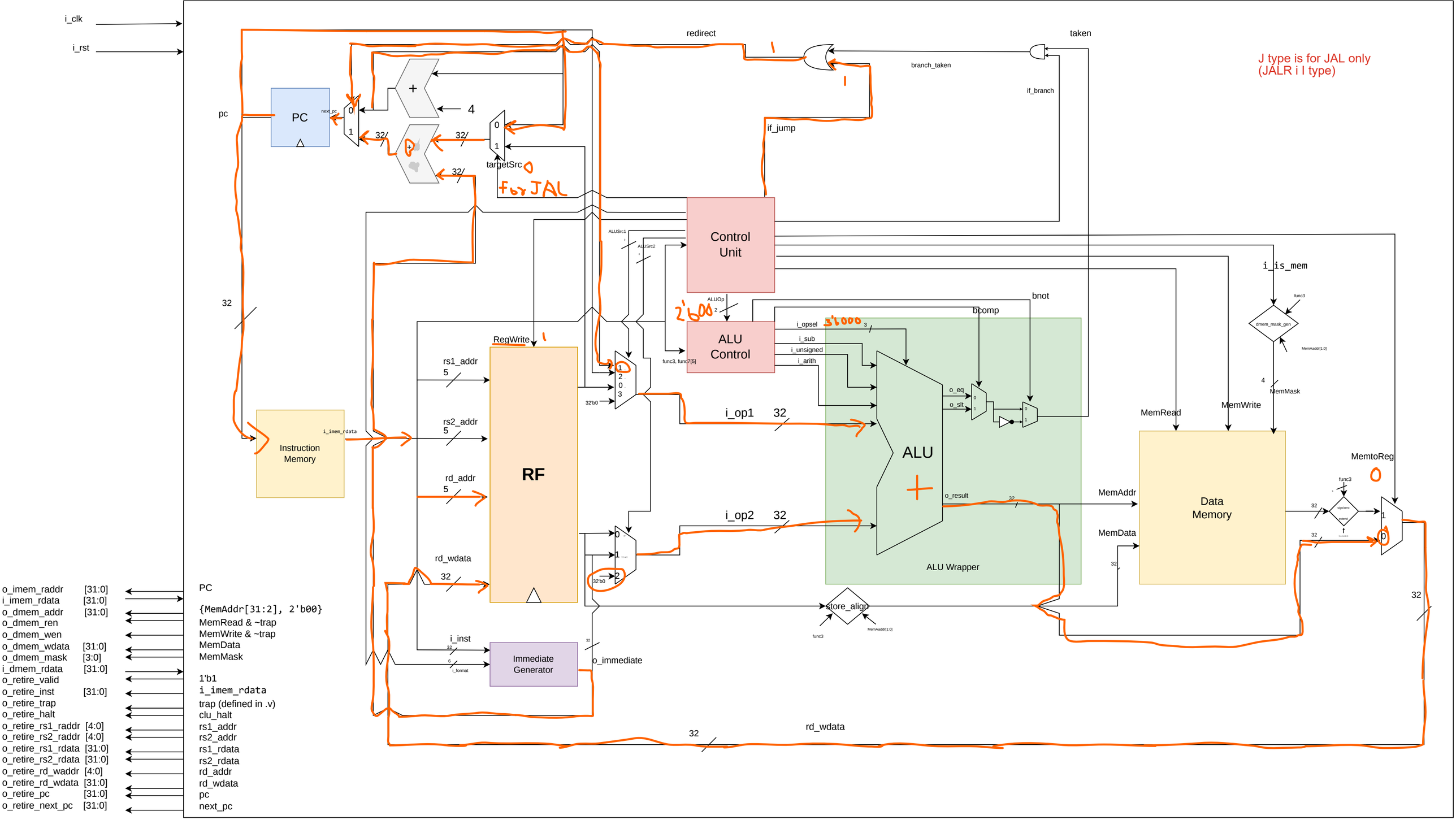

i_clk
i_rst
redirect
taken
branch_taken
if_branch
J type is for JAL only
(JALR i I type)
pc
PC
next_pc
4
32
targetSrc
for JAL
if_jump
Control
Unit
ALUSrc1
ALUSrc2
ALUOp
2'b00
bnot
bcomp
i_opsel
5'b000
i_sub
i_unsigned
i_arith
func3, func7[5]
ALU
Control
RegWrite
i_is_mem
func3
dmem_mask_gen
MemAddr[1:0]
4
MemMask
MemRead
MemWrite
MemtoReg
rs1_addr
5
rs2_addr
5
rd_addr
5
RF
rd_wdata
i_op1
i_op2
ALU
o_eq
o_slt
o_result
ALU Wrapper
MemAddr
MemData
Data
Memory
signExtend
zeroExtend
Instruction
Memory
i_imem_rdata
32'b0
i_inst
i_format
Immediate
Generator
o_immediate
store_align
func3
MemAddr[1:0]
o_imem_raddr [31:0]
i_imem_rdata [31:0]
o_dmem_addr [31:0]
o_dmem_ren
o_dmem_wen
o_dmem_wdata [31:0]
o_dmem_mask [3:0]
i_dmem_rdata [31:0]
o_retire_valid
o_retire_inst [31:0]
o_retire_trap
o_retire_halt
o_retire_rs1_raddr [4:0]
o_retire_rs2_raddr [4:0]
o_retire_rs1_rdata [31:0]
o_retire_rs2_rdata [31:0]
o_retire_rd_waddr [4:0]
o_retire_rd_wdata [31:0]
o_retire_pc [31:0]
o_retire_next_pc [31:0]
PC
{MemAddr[31:2], 2'b00}
MemRead & ~trap
MemWrite & ~trap
MemData
MemMask
1'b1
i_imem_rdata
trap (defined in .v)
clu_halt
rs1_addr
rs2_addr
rs1_rdata
rs2_rdata
rd_addr
rd_wdata
pc
next_pc